

An Examination of the Effectiveness and Limitations of Online Decentralized Participation in Social Decision-Making Processes

A Comparative Study of Two local community Issues and Two Types of Decision Frameworks

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Abstract

Recently, the DAO (Decentralized Autonomous Organization), where participants make decisions equally through online voting, has been gaining attention, while traditional methods involve a multi-stage approach, beginning with discussions among stakeholders to extract key issues and followed by evaluations from randomly selected citizens. However, it remains unclear which issues will gain social acceptance under these different methods. This study investigates whether the evaluation of these processes differs depending on the type of issues. A web-based questionnaire survey was conducted using a between-participants two-factor design with a scenario experiment. Participants read scenarios on local community issues (whether to build a geothermal power plant or how to revitalize a shopping district) and decision frameworks (decentralized online or multi-stage decision-making) and responded to related questions. The results indicated that decentralized online struggles with issues involving significant conflicts of interest. However, it is more accepted when applied to relatively low-conflict issues where participants feel empowered.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

Keywords

online distributed, citizen participation, social acceptance, social decision-making process

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1 Introduction

In recent years, various ICT technologies, such as online platforms and mobile applications, have gained attention as a means of promoting citizen participation [1]. In terms of citizen participation, a deliberative democratic role is expected [2], such as fostering community building and forming public opinion to address social issues. However, traditional citizen participation often involves people gathering in the same place for discussions, which can sometimes extend over a long period of time [3]. Such methods often limit participation to specific attitudes, making it hard for time-constrained individuals to join, which can lead to biased opinions. To address these issues, recent studies have explored the development of online tools that gather opinions from a wide range of people, including highly interested individuals from outside the region, and facilitate discussions. Additionally, studies on the impact of these tools on people have been carried out in fields such as HCI, collective intelligence, and AI [4–6].

One such effort gaining attention is the DAO (Decentralized Autonomous Organization), which is recognized as a new online collective and democratic approach based on the core principles of decentralization, automated, and autonomous [7]. It operates by linking with NFTs and tokens, which are types of cryptocurrency assets, and people primarily form online groups to collaborate toward common goals. In DAOs, everyone can participate equally in decision-making through a voting system without relying on a centralized structure. A growing movement aims to use this new collective to address local issues, with high expectations for its impact. For example, in 2021, the Yamakoshi DAO was launched

in a former Niigata village, where “digital residents” worldwide support through funding and democratic decisions [8].

Despite the promise of such online mechanisms, there may be insufficient consideration toward whether directly affected residents accept these mechanisms when addressing issues within local communities. The social acceptance and acceptance of decision-making process is crucial in new technologies or social issues [9–11]. Therefore, it is necessary to clarify the key considerations for the design and implementation required for such technological support to ensure its acceptance. However, empirical studies on which types of local community issues these online mechanisms can be effectively applied to remain insufficient. Therefore, this study aims to provide empirical insights into the aspects that should be considered when designing systems by comparing traditional decision-making methods with the new approaches enabled by online mechanisms for addressing local community issues.

The contributions of this study to the HCI community are as follows. First, it provides empirical insights into applying online mechanisms to local community issues by comparing social decision-making frameworks in scenario-based experiments. Second, it offers implications for participatory design frameworks of social decision-making that are more inclusively accepted by integrating technological advancements with social science perspectives.

2 Related WORK

2.1 Framework of social decision-making

In regional social decision-making, a multi-stage approach [12], such as hybrid-type conferences [13, 14], has traditionally been adopted. This approach involves a two-step process: stakeholder conferences, where individuals with strong interests or stakes in an issue discuss and identify key points, followed by citizen panel conferences, where randomly selected citizens evaluate the issue from a public perspective. It is particularly recommended in cases where opinions are divided on a given issue and the trust and legitimacy of authoritative decisions are at risk, emphasizing the importance of public participation in consensus-building [15] such as the construction of power plants or waste treatment facilities.

This raises critical concerns about the validity of applying online distributed decision frameworks such as DAOs, which include external participants, especially in contexts where the need for deliberation is significant. In issues involving significant conflicts of interests and values, it is uncertain whether such approach can be accepted, especially by residents.

On the basis of the above, two decision frameworks and two types of local community issues are established. The two frameworks are the online decentralized approach, where anyone with an interest can freely participate and make decisions through voting, and the multi-stage approach represented by hybrid conferences. The two types of local community issues are those likely to involve significant conflicts of interest and values and those where conflicts are relatively minor. This study explores the potential and limitations of online decentralized decision frameworks like DAOs—specifically the conditions under which this approach is suitable or unsuitable—by comparing evaluations when the two decision-making frameworks are applied to different local community issue scenarios.

2.2 Psychological factors in evaluating social decision-making process

In research on such social decision-making processes and social acceptance, the following psychological scales are used for evaluation. The key concepts that related studies focus on are legitimacy, procedural fairness, and competence.

In decision-making related to shared community resources, legitimacy, which evaluates the degree of approval on the basis of rights and appropriateness, is a critical factor. An approval model based on Feather’s legitimacy model [16] has been proposed [17], suggesting that the determinants of legitimacy include institutional substance—criteria for judging rights on the basis of legal, political, or social norms—and perceived substance, which involves judgments of rights based on evaluations such as desirability, favorability, and trust. Additionally, resentment, which may arise from the absence of these foundations, are also hypothesized to influence perceptions of legitimacy. These factors are also believed to influence the acceptance of the decision-making procedure itself as a valid approach.

In social decision-making processes, including citizen participation, procedural fairness—defined by factors such as transparency and the ethicality of the decision-making process—is crucial [18, 19]. Procedural fairness has several antecedent factors [20, 21], such as information disclosure and the reflection of voice. Additionally, there are factors such as authority control, which ensures that decision-making is not dominated by individuals with significant influence; bias suppression, which aims to prevent decisions from being skewed toward specific interests or viewpoints; and representativeness, which examines whether the participants accurately reflect the broader population. There is also inclusiveness, which emphasizes the importance of incorporating diverse perspectives, ensuring that both majority and minority voices are considered in the decision-making process.

Moreover, the aspect of competence has also been highlighted as an important evaluation criterion for citizen participation [21]. Competence can be assessed by measuring factors such as empowerment, which refers to perceived benefits participants may gain from their participation, and costs associated with participation, such as time and effort [22].

Based on the above, this study aims to examine whether the evaluation of these factors and acceptance of the decision framework, and the factors influencing this acceptance differ depending on the combination of decision frameworks applied to two local community issue scenarios. While the openness and decentralized nature of DAOs allows anyone with an interest to participate, this inclusiveness raises questions of legitimacy and fairness when decisions involve resource sharing and the bearing of risks specific to a particular community. It is possible that for local issues involving significant conflicts of interest and values, online decentralized approaches may have lower legitimacy, procedural fairness, and acceptance. This study aims to clarify this prediction, and what factors influence acceptance in online conditions.

3 Method

3.1 Outline of experiment

The scenario experiment was conducted online from March 2 to 10, 2023 by Cross Marketing Inc. Approximately 1,000 participants took part, ensuring equal representation across conditions (4: two types of local community issues $\times$ two decision-making frameworks), gender (2: man or woman), residential region (6: divided into six blocks across Japan), and age group (6: ten-year increments from 20s to 60s, and 70s and above). The experiment yielded 3,930 valid responses.

The participants, acting as residents of a fictional community, read scenarios about local community issues and corresponding decision frameworks for addressing them. The local issues consisted of two types: construction of a geothermal power plant and revitalization of a shopping district. The frameworks included “distributed online” and “multi-stage” approaches. The experiment was conducted using a two-factor between-subjects design. After reading the scenarios, participants answered a questionnaire. The experiment was conducted with the approval of the ethics review committee of the authors’ affiliated organization. Only those who consented to the handling of their data participated, and they were allowed to stop responding at any point during the experiment.

3.2 Scenario

First, participants were instructed that they lived in a fictional local region known for its hot springs but faced several local community issues, such as a declining population over the years.

The scenarios related to these issues included whether or not to construct a geothermal power plant (hereafter referred to as Issue A) and revitalizing a shopping district (hereafter referred to as Issue B). The Issue A scenario is considered to likely involve significant conflicts of interest and values, while the Issue B scenario is regarded as involving relatively minor conflict. The Issue A scenario involved a plan to construct a power plant that was expected to provide clean and stable energy supply; however, despite safety assessments, similar constructions in other regions had altered the quality of the hot springs, resulting in a decline in visitors or even the depletion of the springs. The Issue B scenario involved that there were no young successors to inherit historic inns in the area and that construction of large hotels had further reduced foot traffic to the shopping district, so they needed to develop revitalization plans to restore the vibrancy of the shopping district.

The decision framework scenarios included “distributed online” (hereafter referred to as online) and “multi-stage decision-making” (hereafter referred to as multi-stage). In the online scenario, people from inside and outside the region with an interest or concern gathered online, and any group decision was made through equal online voting by all members. In the multi-stage scenario, participants were told that the decision-making process involved an initial meeting where stakeholders and highly concerned individuals identified key discussion points, followed by another meeting where randomly selected citizens evaluated those points from the general public’s perspective to reach a decision.

Participants were randomly assigned to one of the four combinations of local community issues and decision frameworks. After

reading the scenarios, they were instructed to respond to a questionnaire regarding the proposed approach as if they were residents of the local region. The full text of each scenario can be found in Appendix H2.

3.3 Questionnaire

The questionnaire included 14 psychological scales (Appendix H1), demographics, and other related information.

First, regarding legitimacy, institutional substance, perceived substance, and resentment were measured with two items as antecedent factors of legitimacy, such as “I somehow feel dissatisfied with the involvement of those people” (resentment). Legitimacy was measured with three items to ask whether participants could approve the involvement of the people described in the decision framework scenario. Next, regarding procedural fairness, reflection of voice, information disclosure, authority control, bias-suppression, representativeness, and inclusiveness were measured with two items as antecedent factors of procedural fairness. For example, information disclosure (reversed items) was measured using “The necessary information for decision-making will likely not be fully disclosed,” and representativeness (reversed items) was measured using “The participants are likely limited to those with a high level of interest, and their opinions may not represent those of all individuals affected by the issue.” Procedural fairness was measured with two items asking for the fairness of the entire decision process to be evaluated. Then, regarding competence, empowerment was measured with four items, such as “Participating in this approach can broaden the perspectives of the participants,” and cost was measured with two items, each regarding effort and time. Finally, acceptance was measured with two items to ask whether participants could accept the decision-making approach. All items were measured using a 5-point Likert-scale ranging from 1 (do not agree at all) to 5 (agree completely), and reversed items were adjusted through reverse processing and used in subsequent analyses. The standardized reliability coefficients (Cronbach α) generally indicated sufficient reliability (α s $> .81$), though some scales were lower, such as institutional substance ($\alpha = .78$), perceived substance ($\alpha = .47$), and inclusiveness ($\alpha = .65$). However, these psychological constructs were developed as coherent concept based on prior research [17, 23] and were used as single scales in the subsequent analyses.

Items related to demographics and other related information of the participants are described below. First, participants were asked to subjectively evaluate the rural-to-urban scale of their residential area on an 11-point scale (hereafter referred to as “subjective urbanity”) [24]. Participants were also asked about the activities they engage in outside of work (10 items for example activities related to health and medical services, people with disabilities, and so on) [25] using a 5-point scale from “1: Never” to “5: Almost every day.” A dummy variable was created to indicate whether participants had never engaged in these activities (total score of 10) or had engaged in them (total score greater than 10). Participants were asked about the frequency of their SNS usage with five items (e.g., “checking social networking sites such as Facebook, Twitter, or Instagram”) [26]. They responded on a 5-point scale from “1: Never” to “5: Almost every day.” On the basis of the distribution of responses, participants with a total score of 5 were categorized

Table 1: ANOVA

Scale	Issue type	Decision framework: Online	Decision framework: Multi-stage	Main effect of issue		Main effect of decision framework		Interaction	
		M (SD)	M (SD)	F	η^2_p	F	η^2_p	F	η^2_p
Institutional substance	Issue A	3.27 (0.91)	3.43 (0.89)	160.37 ***	0.04	26.52 ***	0.01	0.17	0.00
	Issue B	2.93 (0.89)	3.06 (0.86)						
Perceived substance	Issue A	2.98 (0.74)	3.05 (0.72)	19.77 ***	0.01	0.11	0.00	7.79 **	0.00
	Issue B	3.15 (0.68)	3.09 (0.72)						
Resentment	Issue A	2.64 (0.91)	2.46 (0.86)	46.24 ***	0.01	7.53 **	0.00	14.27 ***	0.00
	Issue B	2.34 (0.88)	2.37 (0.87)						
Legitimacy	Issue A	3.47 (0.85)	3.62 (0.84)	30.22 ***	0.01	12.36 ***	0.00	5.39 *	0.00
	Issue B	3.67 (0.84)	3.71 (0.78)						
Reflection of voice	Issue A	3.32 (0.80)	3.33 (0.81)	26.84 ***	0.01	0.19	0.000	0.58	0.00
	Issue B	3.47 (0.75)	3.44 (0.78)						
Information disclosure	Issue A	2.91 (0.83)	2.92 (0.85)	72.22 ***	0.02	0.87	0.00	1.51	0.00
	Issue B	3.16 (0.78)	3.10 (0.77)						
Authority control	Issue A	3.20 (0.90)	3.31 (0.86)	22.09***	0.01	54.62 ***	0.01	12.16 ***	0.00
	Issue B	2.97 (0.89)	3.28 (0.84)						
Bias- suppression	Issue A	3.03 (0.88)	3.03 (0.84)	35.62 ***	0.01	3.33 [†]	0.00	4.06 *	0.00
	Issue B	3.24 (0.78)	3.14 (0.81)						
Representativeness	Issue A	2.33 (0.81)	2.62 (0.85)	38.05 ***	0.01	48.70 ***	0.01	13.94 ***	0.00
	Issue B	2.59 (0.81)	2.68 (0.83)						
Inclusiveness	Issue A	3.08 (0.75)	3.11 (0.77)	35.14 ***	0.01	0.90	0.00	4.40*	0.00
	Issue B	3.28 (0.72)	3.20 (0.76)						
Procedural fairness	Issue A	3.19 (0.88)	3.37 (0.83)	36.64 ***	0.01	14.90 ***	0.00	10.77 *	0.00
	Issue B	3.43 (0.78)	3.45 (0.81)						
Empowerment	Issue A	3.57 (0.70)	3.59 (0.70)	14.27 ***	0.00	1.51	0.00	4.71 *	0.00
	Issue B	3.70 (0.66)	3.63 (0.70)						
Cost	Issue A	3.65 (0.84)	4.00 (0.76)	0.02	0.00	168.75 ***	0.04	1.29	0.00
	Issue B	3.67 (0.82)	3.97 (0.72)						
Acceptance	Issue A	3.21 (0.93)	3.42 (0.86)	54.83 ***	0.01	13.38 ***	0.00	16.13 ***	0.00
	Issue B	3.52 (0.82)	3.51 (0.80)						

^a ***p <.001, **p<.01, *p<.05, [†]p<.10

as “no use,” those with a total score of 12 were categorized as “low use,” and those with a total score greater than 12 were categorized as “active use.”

4 Result

4.1 Comparison of mean values for each scale between conditions

The results of the two-way Analysis of Variance (ANOVA) for the psychological scales are shown in Table 1. The result revealed that for institutional substance, the main effect of local community issue and decision framework were significant, but the interaction effect was not significant. Main effects of the local community issue were observed for reflection of voice and information disclosure. Specifically, the issue related to the geothermal power plant was evaluated as less likely to reflect opinions and more likely to conceal information. On the other hand, the only significant main effect of the decision framework was observed for cost. This indicates that the distributed online framework was evaluated as having

lower time and effort costs for participation. For the other 10 scales, significant interaction effects were observed. Therefore, the results of the multiple comparison analysis are presented in Figure 1.

The most notable differences were observed in the “Issue A × online” condition, where resentment was higher, and legitimacy was lower compared to with the other three conditions. Additionally, representation and inclusiveness were rated lower, as were procedural fairness and acceptance. In other words, implementing an “online” decision framework for issues such as the geothermal power plant may not be approved, the process may be perceived as unfair, and it may not be accepted. On the other hand, in the “Issue B × online” condition, authority control was shown to be lower, and bias suppression was higher compared to with the other three conditions. This indicated that a decision framework like the “online” approach was evaluated as less centralized and less likely to favor specific interests or values, particularly in the context of the shopping district revitalization issue. Furthermore, the “Issue B × online” condition showed higher results for perceived

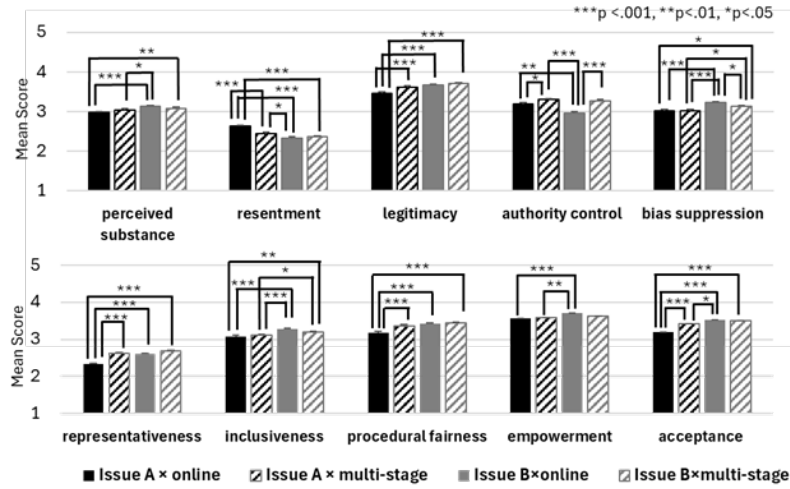


Figure 1: Results of multiple comparison analysis (Error bars represent SE)

Table 2: : Multiple regression analysis (Acceptance, BIC)

Scale	Issue A × online			Issue A × multi-stage			Issue B × online			Issue B × multi-stage		
	Estimate	Std. error	t	Estimate	Std. error	t	Estimate	Std. error	t	Estimate	Std. error	t
Intercept	0.44	0.12	3.53***	-0.21	0.08	-2.65**	0.18	0.13	1.38	0.06	0.08	0.667 [†]
Perceived substance	0.09	0.02	3.50***									
Resentment	-0.14	0.02	-7.74***				-0.06	0.02	-2.98**			
Legitimacy				0.15	0.01	7.11***	0.12	0.02	5.27***	0.15	0.02	6.77***
Reflection of voice				0.08	0.02	3.45***	0.10	0.03	3.77***			
Bias-suppression										0.07	0.02	3.32***
Inclusiveness	0.11	0.03	4.20***	0.08	0.02	3.40***						
Procedural fairness	0.67	0.03	26.29***	0.61	0.03	23.59***	0.64	0.03	24.41***	0.60	0.02	24.48***
Empowerment	0.11	0.03	3.94***	0.14	0.03	4.65***	0.14	0.03	5.00***	0.17	0.03	6.24***
Adjusted R-squared	0.73			0.74			0.72			0.70		
F	521.5 (5, 976) ***			568.1 (5, 976) ***			502.3 (3, 977) ***			583.3 (4, 978) ***		

a ***p < .001, **p < .01, *p < .05, [†]p < .10

substance, resentment, acceptance, and empowerment compared to with both the “Issue A × online” and “Issue A × multi-stage” conditions. Introducing a distributed online approach for issues like shopping district revitalization, rather than the geothermal power plant, was evaluated as a process that could be entrusted to the group involved, and negative emotional responses were lower. Additionally, the high empowerment scores suggested that the perceived benefits of participation were expected to be significant, and the decision-making method was also evaluated as more acceptable.

4.2 Factors influencing acceptance: multiple regression analysis by conditions.

Multiple regression analyses were conducted for each condition, with acceptance as the dependent variable. The independent variables included the other 13 psychological scales, as well as gender, age group, subjective urbanity, participation in activities, and SNS usage frequency. In this study, because of the large sample size, we used the Bayesian Information Criterion (BIC) as the criterion for selecting the optimal model.

The best-fit model contained procedural fairness and empowerment across all conditions, but it did not contain some factors such as institutional substance, gender, age, participation in activities, and SNS usage frequency under any condition. For “Issue A,” inclusiveness—whether diverse values and minority opinions were included—was contained in the best-fit model for both decision frameworks. Next, focusing on the decision framework, for “online,” resentment was included in the best-fit model as a negative effect of acceptance, regardless of the issue. Then, when focusing on each condition, the “Issue A $\times$ online” condition included the perceived substance, indicating that whether the involved individuals are deemed appropriate is important for acceptance. Legitimacy was not included in this condition, although it was included in the other three conditions. In the “Issue B $\times$ multi-stage” condition, bias-suppression was included only in this condition, suggesting that the absence of bias toward specific interests or perspectives influences acceptance.

5 Discussion

This study explores how online citizen participation and its support methods can gain social acceptance from residents while considering legitimacy and fairness, and to identify which local community issues are best suited for such approaches. In this study, two types of issues were set for examination: the “geothermal power plant construction” issue, which requires deliberation, and the “shopping district revitalization” issue, which involves relatively low conflicts of interest or values. A two-factor experimental design with scenarios was used to evaluate how decision-making frameworks are assessed. The results showed that for the “geothermal power plant construction” issue, the online method was evaluated lower in factors of legitimacy, procedural fairness, and acceptance compared with the other three conditions. In the “online” condition for both issues, resentment was shown to have a negative impact on acceptance. For the geothermal power plant issue, the “online” approach triggered a stronger resentment compared with the other three conditions, suggesting that resentment could be a barrier to acceptance. On the other hand, for issues like the “shopping district revitalization” issue, it was rated similarly to the multi-stage approach, suggesting that the online method may be applicable depending on the type of issue. While the traditional multi-stage approach appeared to foster acceptance because it seemed unbiased toward any particular interests or viewpoints, the online distributed approach was positively evaluated for allowing anyone with an interest to participate at a low cost and for avoiding the influence of dominant voices. The result also suggest that the online approach could be applicable to issues where participants can expect their contributions to have an impact and feel effective.

Additionally, institutional substance was not contained in the best-fit model for any condition as a factor influencing acceptance. Moreover, acceptance was not influenced by individual attributes such as frequency of SNS usage, participation in activities, age, or gender. For example, a higher SNS usage frequency did not necessarily lead to acceptance in the online method. These results suggest that when adopting an online decision framework, it is important to focus on aspects like procedural fairness and to carefully consider which types of issues are suitable for such methods.

However, local community issues are diverse, and the surrounding conditions are not uniform. Therefore, it is difficult to clearly determine which types of issues are suitable for these methods on the basis of only this study. Even in the case of shopping district revitalization, some situations may involve serious conflicts of interest, and some people may believe that decisions should be made only by highly interested individuals rather than randomly selected citizens. Given the small effect size, it is also important to be cautious when interpreting the results. Moreover, it is also necessary to clarify the factors that influence not only acceptance but also people’s willingness to actively participate in the process.

Despite these limitations, this study found that the online decentralized approach may be applicable to issues like shopping district revitalization rather than issues like the construction of a geothermal power plant, which involves significant conflicts of interest and values, as they require deliberation by a diverse range of people with little interest in the issue. To implement an online decentralized approach, it may be important to focus on designing participation processes that highlight the benefits participants can gain, such as exposure to diverse values and perspectives and the expansion of social networks. Additionally, it is necessary to identify the types of issues and their characteristics that can better provide these benefits. This study demonstrated the significance of examining online decentralized groups and decision-making methods, such as DAOs, from not only technological advancements and innovative aspects but also a social science approach. In the future, it may be necessary to design a consensus-building process that combines the multi-stage approach and the online decentralized approach utilizing the advantages of both methods by collaborating across disciplines.

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